Lena Varga-Holm

Senior Data Engineer | Data Platform

PROFESSIONAL SUMMARY

Senior Data Engineer with experience building and operating production data platforms, cloud data warehouses, streaming ingestion, and data pipelines. Focused on reliable data delivery, data quality, and platform operations across Airflow, dbt, Kafka, Snowflake, Python, and SQL. Has delivered high-volume event pipelines, warehouse transformation improvements, and production database load reductions.

EXPERIENCE

Senior Data Engineer | Lukovac Freight Hub d.o.o. | 2022–Present

- Owns ingestion and orchestration for the data platform using Airflow, Snowflake, dbt, and Kafka.
- Participates in the data platform on-call rotation, one week in six.
- Mentors two mid-level engineers and runs data platform office hours.
- Reduced the Snowflake bill from approximately 38k/month to 24k/month in 2024 through warehouse sizing, auto-suspend, and removal of unused tables.

Data Engineer | Kovalent Pay | 2019–2022

- Built PySpark workloads on AWS EMR, designed a Redshift warehouse, and implemented data quality checks with Great Expectations.

SELECTED PROJECTS

Change Data Capture from Postgres | Lukovac Freight Hub | 2024

- Replaced hourly full-table extracts of the orders database with change data capture using Debezium into Kafka.
- Reduced replica CPU during loads from approximately 70% to 25%.
- Technologies: Debezium, Kafka, PostgreSQL, Airflow, Snowflake.

Shipment Event Streaming Pipeline | Lukovac Freight Hub | 2023

- Designed and built a streaming pipeline for shipment tracking events from courier partners, processing around 30 million events per day with end-to-end latency under two minutes at p95.
- Routed partner webhooks into Kafka topics and used a Kafka Connect sink to Snowflake via Snowpipe Streaming.
- Added a dead-letter topic and replay tooling so failed events could be reprocessed without data loss.
- Technologies: Kafka, Kafka Connect, Snowflake, Python, Terraform, AWS MSK.

Open-Source Contribution | 2023

- Fixed a pagination bug and added tests to a community Airflow provider for a SaaS CRM API; two pull requests were merged.
- Technologies: Python, Airflow, pytest.

Migration of the Warehouse to dbt | Lukovac Freight Hub | 2022–2023

- Migrated approximately 300 stored procedures into dbt models with tests and documentation.
- Introduced incremental models and slim CI using state:modified in GitHub Actions.
- Cut the nightly warehouse build from 5 hours 10 minutes to 1 hour 35 minutes and made lineage visible to analysts.
- Technologies: dbt, Snowflake, SQL, GitHub Actions.

Transaction Reconciliation Engine | Kovalent Pay | 2020–2021

- Built PySpark batch jobs on EMR to match bank settlement files against internal ledgers.
- Reduced the finance team's manual reconciliation work from approximately three days to four hours per month-end.
- Technologies: PySpark, AWS EMR, S3, Glue Data Catalog, Redshift, Great Expectations.

SKILLS

Python, SQL, PySpark, Airflow, dbt, Kafka, Kafka Connect, Debezium, Snowflake, Snowpipe Streaming, Amazon Web Services (EMR, MSK, S3), Redshift, PostgreSQL, Great Expectations, Terraform, GitHub Actions, pytest, data quality, data orchestration, change data capture, streaming data pipelines, cloud data warehouses.

CERTIFICATION

AWS Certified Data Engineer – Associate | May 2024

EDUCATION

MSc in Computer Science | Northgate Technical University
Thesis: Incremental materialized views.

COMMUNITY

- Teach SQL for beginners at a local women-in-tech meetup, approximately four evenings per year since 2021.
- Presented "Data contracts without the hype" at a regional data meetup in 2024.